

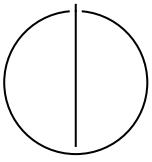
SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
CHAIR OF HEALTH INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Decoding Music Perception: EEG-Based Genre Classification: Knowledge Base

Lukas Alex Jochim





SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — CHAIR OF HEALTH INFORMATICS

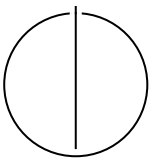
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Decoding Music Perception: EEG-Based Genre Classification: Knowledge Base

Dekodierung der Musikwahrnehmung: EEG-basierte Genre-Klassifizierung

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I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, 01.09.2025 Lukas Alex Jochim

Acknowledgments

Abstract

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1 Introduction

1.1 Motivation

The human brain is one of the most complex and important systems within the human body. The brain manages various functionalities like emotion, memory, attention and sensory processing, to name a few. Apart from that, the brain plays a significant role in how we process musical pieces. Independent of culture and age, music is a universal tool to influence one's emotions, concentration and cognition. Furthermore, the brain gives us the unique ability to retrieve many different features out of music, like melody extraction or the identification of an artist or a specific genre. Depending on the style of music, the brain reacts differently. However, although the brain is central to cognitive processes such as listening to music, the core mechanisms remain unknown and are currently black boxes.

To enlighten these black boxes, medical advancements like functional magnetic resonance imaging (fMRI) or electroencephalogram (EEG) arrived, which can measure brain activity. Especially EEG has aroused high interest since they directly measure the neural activity and can be done non-invasive with a little overhead in setup. This is why combining EEG and music has the ability to help better understand how our brain functions. By further adding means of Machine Learning (ML), a whole new interdisciplinary field opens up, bridging the gap between neurosciences and Computer science.

There is a growing research interest in the affective influence of music on the brain. However, classifying different music genres based on EEG signals remains underrepresented. This is exactly where this thesis aims to close the gap, by investigating whether it is possible to classify various musical genres depending on the EEG signals acquired using state-of-the-art ML techniques.

The research results have high potential in the domain of medical computer science. Especially with the growing investigations on Brain Machine interfaces (BMIs), adaptive music therapy, personalised music recommendations or neurofeedback systems may become possible. Additionally, entertainment technologies can become more user-centred, facilitating unique experiences that impact many sectors like education and mental health

1.2 Research goals

1.3 Methodology

2 Theoretical Background

2.1 Medical foundations of EEG signals

2.2 Brain rythms

2.3 Correlation between music and brain activity

2.4 Recording and measurement of EEG signals

2.5 Denoising and preprocessing of EEG signals

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3 Related work

3.1 Music Genre classification with audio features

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3.2.1 DEAP dataset

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4 Music datasets

4.1 Audio Set

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4.4 FMA dataset

4.5 GTZAN Genre classification

4.6 Final decision

5 Methodology

5.1 Acquisition

5.1.1 General setup, environment and stimuli

The experiment has been conducted in a laboratory setting. As mentioned in Chapter 4, the GTZAN Music dataset is chosen as a basis for the music stimuli. However, for the sake of the quality of the data, it would not be beneficial to hear all 1000 songs in one session. To avoid quick attention loss during a session and gather as much different data under different circumstances, the whole set of audio files is partitioned into 20 distinct sessions, each lasting around 30 minutes. This means that per session, 50 songs are played. [1] For each of these songs, not only are EEG signals measured, but also other biosignals like PPG, IMU and temperature. However, this work will solely focus on the recorded EEG data. However, these 50 songs are not chosen randomly. Five songs from each genre are being played during the session to ensure that each of the ten genres from the dataset is represented in each session. This also ensures no genre bias in the data, thus resulting in a balanced signal across genres. During listening, the order of the songs is randomised to evoke a brain reaction that is as natural as possible, as the brain cannot predict the genre that is being listened to. Additionally, a five-second break is implemented in between songs to not overstimulate the brain.

All 20 sessions have been recorded over two weeks, with an average of two sessions per day. Between each session, there was a short break of between ten and 30 minutes. This verifies a sufficient pause of the brain to fully prepare for the next session. The test person is the author, a 21-year-old, right-handed college student without any common (brain) diseases. To further guarantee a high quality in the acquired signal data, the experiment's environment and setting must be adapted. First and foremost, the test person is instructed to sit comfortably in a chair with his eyes closed to minimise artificats coming from blinking and eye-movement. Furthermore and most importantly, any movements, especially in the head region, must be limited and as minimal as possible since the EEG device reacts very sensitive to head movements. Apart from that, the room was quiet and isolated with dimmed natural lighting combined with the

[1] Because of a corrupt audio file, one session only consists of 49 songs

brightness from a desktop monitor. The songs were played over the EEG headset (cf. **??**) with average and constant volume.

5.1.2 Acquisition device

A non-traditional approach is chosen to record the EEG signal data coming from the brain. For this study, a prototype for a hybrid biosensing platform *Hugin* by Jabra is chosen. It is a Bluetooth headset that can play music and measure biosignals at the same time. For Audio, it uses classic Bluetooth to connect to a standard PC or Laptop. For the data, however, it uses a second low-energy bluetooth channel for the biosignal data, which can be connected to a second data acquisition PC.

The headphones have multiple sensors and electrodes to measure biosignals like EEG, PPG, IMU and temperature. For EEG, the headphones are supplied with 2 × 4 sensor electrodes and two bias electrodes. The sensors are easily configurable and provide an ultra-low-noise front-end, meaning that the data points are amplified with minimal distortion and interference. Besides that, the electrodes sample with 24-bit and a frequency of 250Hz.

For PPG, one optical sensor on the right side provides a high resolution of 19-bit ADC with a configurable sample rate. They also support raw data output or built-in algorithms to conclude heart rate or SpO2.

The IMU data is sensed by a 3-axis digital accelerometer fused with a 3-axis digital gyroscope. On a full scale, the accelerometer can recognize acceleration from $\pm 2g$ up to $\pm 16g$ (in steps of the power of two). The gyroscope can measure angular velocity from $\pm 125°/s$ up to $\pm 4000°/s$. The recognised data is then combined to provide accurate information on orientation, movements and position.

Last but not least, there are three digital thermometers, one on the skin, another one inside the earcup and one on the outside of the earcup to measure ambient temperature. All of those temperature sensors have an accuracy of ± 0.1°C.

Since this headset is still a prototype, they still is ongoing development and research on the headset. Nonetheless, some small-scale research projects already employed the biosignals from the headphones. These projects range from stress detection using the PPG data to device control and the recognition of cognitive fatigue using the EEG sensors.

5.1.3 Technical setup

Multiple technical components have to come together to gather the experiment's data. All the components employ the Lab Streaming Layer (LSL). It provides a standardised

way to stream time-sensitive data between devices and applications. Devices or apps then push data on this stream, which is captured and stored in a special file format.

At the core, there are two main programs responsible for the data. On the one hand, there is the Headset platform called *BioGNee*. This program is responsible for the sensor setup and starting and stopping the signal acquisition. Communicating with the headset's low-energy Bluetooth Channel reliably sets up the biosensors of the hybrid headphones, making the device ready for acquisition. After a successful setup, the data is pushed on the corresponding LSL Stream. On the other hand, the study uses a custom Python script. This Python script has two main tasks. Firstly, it loads all the songs from one of the 20 pre-generated sessions and randomly shuffles them. After that, the program plays each song one by one. Simultaneously, the script also pushes timestamps during recording to assign each segment to each song later.

The LSL Recorder combines both of these components. The LSL recorder collects all open streams on which the device/programs push their data. These are collected and stored in an XML-like file called .xdf. The resulting document, therefore, provides a unified way to not only store the signal data but also the respective time markers. These can be visualised by a custom MATLAB script to inspect the data quality. Important to note is that each distinct session occupies one .xdf file.

For future steps like Denoising or Feature Extraction, it is essential to look at each segment individually, i.e., each song. Therefore, a further custom Python program is used for the segmentation process of one session. Since the start and the end of one song are pushed to the LSL stream, one can read out the time these markers have been pushed and fetch all the EEG data points measured in this interval. These segments are then written in a new .xdf file to allow easy handling in future processing steps and seamlessly work with the existing MATLAB script for EEG visualisation.

5.2 Denoising

5.3 Feature extraction

5.4 Classification

5.4.1 Model 1

5.4.2 Model 2

5.4.3 Model 3

6 Results

6.1 Metric 1

6.2 Metric 2

6.3 Metric 3

7 Conclusion

7.1 Key findings and results

7.2 Problems and Limitations

7.3 Future work

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